laws that *force* the Universe to behave the way it does, that *force* rain to condense on windows, *force* the Sun to burn nuclear fuel, *force* black holes to produce gravitational waves when they collide, and so on.

One might object that each set of laws in the sequence "looks" very different from the preceding set. (For example, the absolute time of Newtonian physics looks very different from the many different time flows of special relativity.) In the "looks" of the laws, there is no sign whatsoever of convergence. Why, then, should we expect convergence? The answer is that one must distinguish sharply between the predictions made by a set of laws and the mental images that the laws convey (what the laws "look like"). I expect convergence only in terms of predictions, but that is all that ultimately counts. The mental images (one absolute time in Newtonian physics versus many time flows in relativistic physics) are not important to the ultimate nature of *reality.* In fact, it is possible to change completely what a set of laws "looks like" without changing its predictions. In Chapter 11, I shall discuss this remarkable fact and give examples, and shall explain its implications for the nature of reality.

Why do I expect convergence in terms of predictions? Because all the evidence we have points to it. Each set of laws has a larger domain of validity than the sets that preceded it: Newton's laws work throughout the domain of everyday life, but not in physicists' particle accelerators and not in exotic parts of the distant Universe, such as pulsars, quasars, and black holes; Einstein's general relativity laws work everywhere in our laboratories, and everywhere in the distant Universe, except deep inside black holes and in the big bang where the Universe was born; the laws of quantum gravity (which we do not yet understand at all well) may turn out to work absolutely everywhere.

Throughout this book, I shall adopt, without apology, the view that there *does* exist an ultimate set of physical laws (which we do not as yet know but which might be quantum gravity), and that those laws truly *do* govern the Universe around us, everywhere. They *force* the Universe to behave the way it does. When I am being extremely accurate, I shall say that the laws we now work with (for example, general relativity) are "an approximation to" or "an approximate description of the true laws. However, I shall usually drop the qualifiers and not distinguish between the true laws and our approximations to them. At these times I shall assert, for example, that "the general relativistic laws [rather than the true laws] *force* a black hole to